Misun Kim

Sheffield, UK · misunkim2732@gmail.com · GitHub · LinkedIn

Research interests

I study why alignment interventions and safety behaviour that work in English fail in other languages, and where in the model the failure arises. My work separates the intervention from the input it acts on, and tests whether automated safety evaluation is equally reliable across languages.

Education

MSc Computer Science with Speech and Natural Language Processing — University of Sheffield
Sheffield, UK · Sep 2025 – Sep 2026 · Distinction (weighted average: 73.17/100)

Dissertation (78/100): Cross-Lingual Activation Steering in Large Language Models: Representational Alignment and Behavioural Transfer

BEng Industrial and Management Engineering; BA Central Asian Studies — Hankuk University of Foreign Studies
South Korea · Mar 2019 – Feb 2024 · GPA 3.94/4.5

Kazakh language and regional studies · First Distinction Scholarship (2023).

Research experience

MSc Dissertation Research — Cross-Lingual Transfer of Activation Steering
University of Sheffield · Supervisors: Dr Cass Zhao and Dr Marco Valentino · Feb 2026 – Sep 2026

- Investigated the robustness of activation-level interventions in multilingual LLMs using Contrastive Activation Addition (CAA), testing whether English-derived steering vectors transfer consistently across three model families and five languages.
- Identified a representation–behaviour gap: high steering-vector similarity did not reliably predict the magnitude or direction of behavioural transfer across models and tasks, concluding that high representational similarity is not sufficient to guarantee comparable behavioural effects, and leaving the question of necessity unresolved.
- Designed layer-wise evaluations with bootstrap confidence intervals, held-out sets, and Gaussian and permutation random-vector baselines to characterise failure modes and distinguish systematic effects from noise.
- Conducted a separate cross-task evaluation using English-derived truthfulness vectors on TruthfulQA, finding a descriptive cross-task contrast across models and behaviours: Qwen3.5-9B transferred sycophancy but not truthfulness, OLMo3-7B-Instruct the reverse.
- Follow-up study (post-MSc): built a full 5×5 source–target transfer matrix with norm-matched controls to distinguish source-vector effects from target-language response differences.

Undergraduate Researcher — ICT Mentoring Project, Ministry of Oceans and Fisheries
Supervisor: Gyu-Young Lee (KAIST) · Mar 2023 – Nov 2023

- First-authored LSTM study on container freight forecasting across five universities, reducing RMSE by 71% over the baseline (published ACK 2023).

Publications & manuscripts

Kim, M., Kim, Y., Kim, E., Lee, B., Han, Y., & Lee, G. (2023). *Implementation of Container Volume Prediction Technology using Deep Learning.* ACK Conference Proceedings, 30(2).

Manuscript in preparation (2027). *Cross-lingual and cross-task generalisation of representation-level interventions in multilingual LLMs.*

Key projects

English–Korean LLM Safety and Judge Validation
Aug 2026 – Nov 2026

- Built a reproducible Python pipeline to evaluate OLMo 3 7B DPO and RLVR checkpoints on 200 matched English–Korean prompts, including 93 harmful items.
- Labelled 70 responses blind to automated verdicts and identified opposing judge errors by language. Human-validated estimates put harmful-request refusal 68–76 percentage points higher in English across the methods and refusal definitions tested.

Multimodal Agent Safety Under Prompt Injection
Nov 2026 – Present

- Preregistered a 200-run evaluation of a Qwen3-VL-8B receipt-processing agent using English and Korean image and text inputs.
- Observed unauthorised mock email calls in 80 of 80 attack runs and zero of 120 controls. The agent still recorded the expense correctly in every attack run.

ESG Compliance Checker
Mar 2026 – Aug 2026

- Built an end-to-end compliance verification system over 11 corporate ESG reports, mapping regulatory criteria to retrieved evidence and producing traceable covered/partial/missing judgements using hybrid retrieval and LLM-assisted verification.
- Benchmarked retrieval and verification separately. Hybrid retrieval with reranking reached 96.3% Hit@5 (26/27 queries), and the verifier matched human labels in 84.6% of fixed-evidence cases (33/39).
- Used retrieval and verifier error analysis to identify annotation gaps and evidence-selection failures, then modularised the system with FastAPI, automated testing, Docker, and a separate Streamlit interface.

Skills

Programming: Python, SQL

ML & NLP: PyTorch, Hugging Face Transformers, scikit-learn

LLM Alignment & Evaluation: Activation steering (CAA), representation analysis, cross-lingual and cross-task evaluation, post-training safety evaluation, LLM-as-judge validation, human annotation and error analysis

Experimental Methods: Controlled experiments, bootstrap confidence intervals, Bayesian uncertainty analysis, random-vector baselines

Research Infrastructure: HPC/SLURM, Weights & Biases, Git, GitHub Actions (CI), Docker, FastAPI, Flask, ChromaDB

Work experience

Sustainability Research Analyst — Eon Consulting Group
South Korea · Mar 2025 – May 2025

- Contributed to 170+ page sustainability reports for major public and financial organisations, conducting TNFD climate scenario analysis and developing a CSDDD-aligned supply-chain risk assessment framework.

Field Service Manager (Data & Operations) — Payhere
South Korea · Mar 2024 – Aug 2024

- Built workflow automation using Slack Workflow Builder, Google Sheets, and Looker Studio to analyse 1,000+ service records, reducing unnecessary field visits by 17%.

Teaching experience

Language Tutor (Private)
Mar 2021 – Nov 2025

- Provided individualized Korean/English instruction to 5+ learners aged 6–18, from early literacy to CSAT-level advanced reading.

Languages

Korean (Native), English (Fluent), Spanish (Intermediate), Kazakh (Intermediate)